

Python Example

<https://github.com/swkim94/cocktail>

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• Summary

- **Python Proficiency:** Demonstrates core programming competency and data engineering depth.
- **API Data Collection:** Fetches and parses raw dataset using a public API.
- **Text Vectorization:** Transforms unstructured data into numeric vectors for analysis.
- **Dimensionality Reduction:** Implements PCA to reduce feature space and compress data.
- **Cluster Separation:** Groups data points cleanly using K-Nearest Neighbors (KNN).
- **Custom Visualizations:** Generates explanatory histograms and PCA-mapped cluster plots.
- **Hardware Automation Note:** my background in instrument automation relies on LabVIEW.
- **Source code:** Fully open-source repository hosted at <https://github.com/swkim94/cocktail>

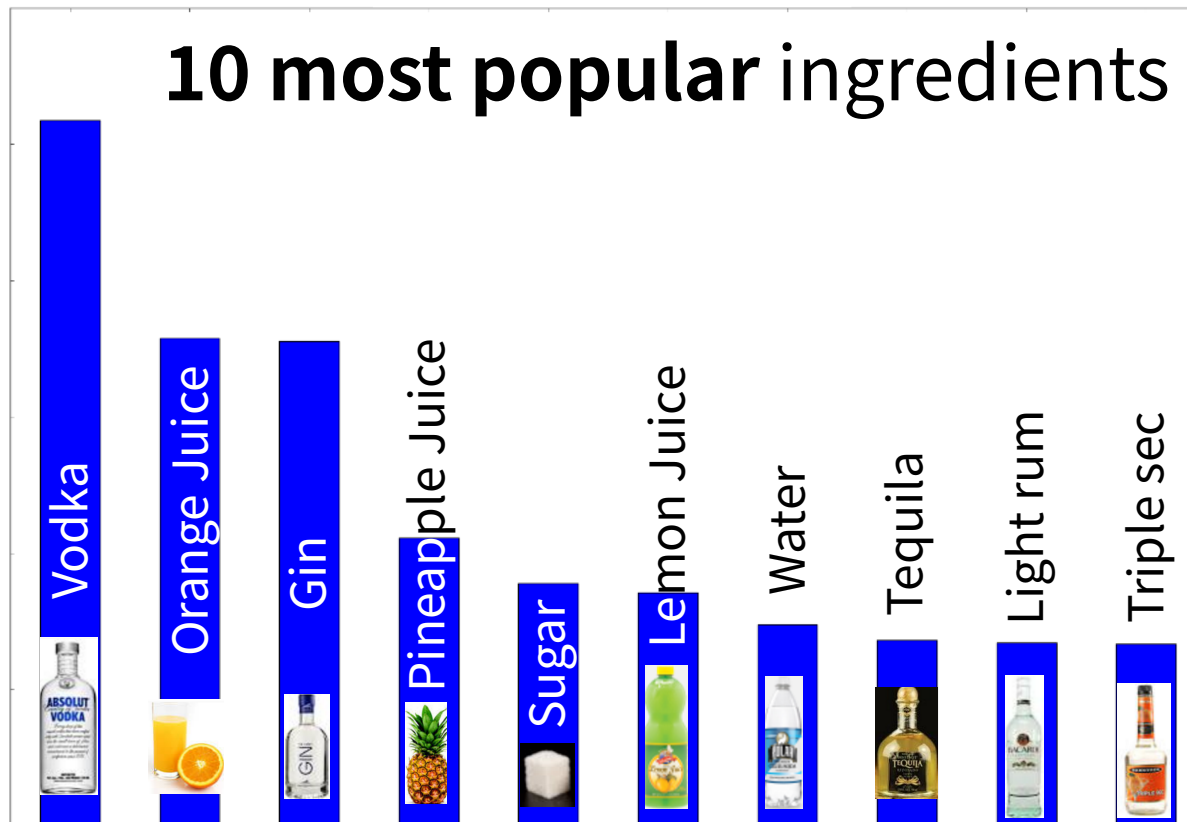
Slide#1: Import Data, Build Pandas Dataframe

TheCocktailDB

<http://www.TheCocktailDB.com>



3081 cocktails
416 ingredients

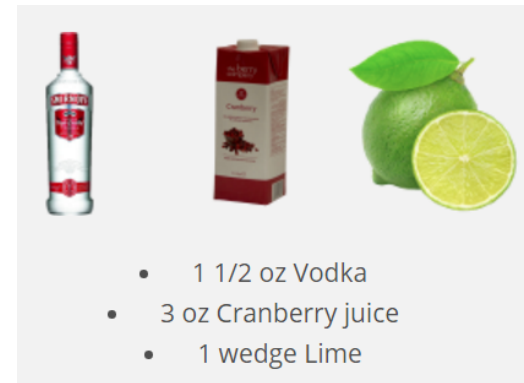
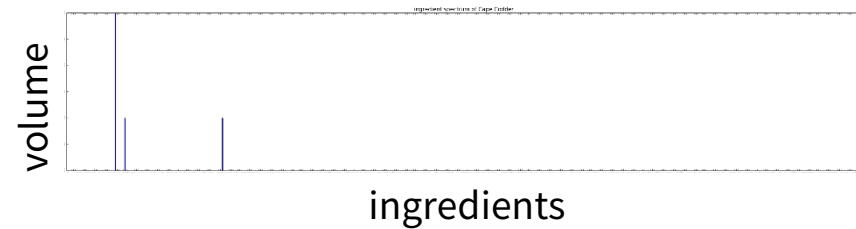
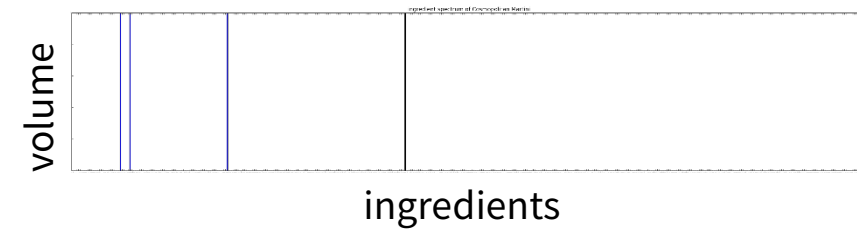


Slide#2: Recommendation (Similarity Search)

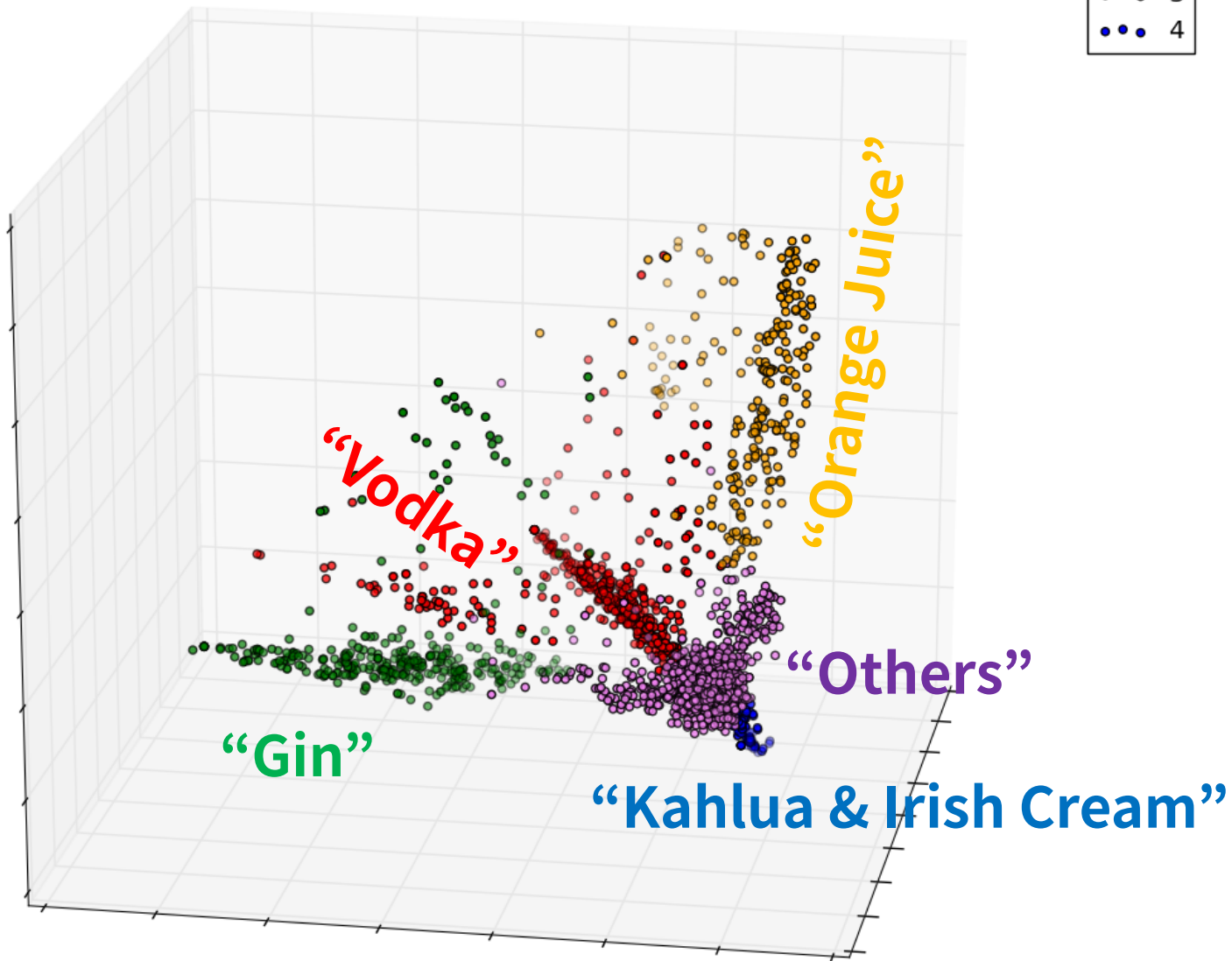
Input: Cosmopolitan Martini



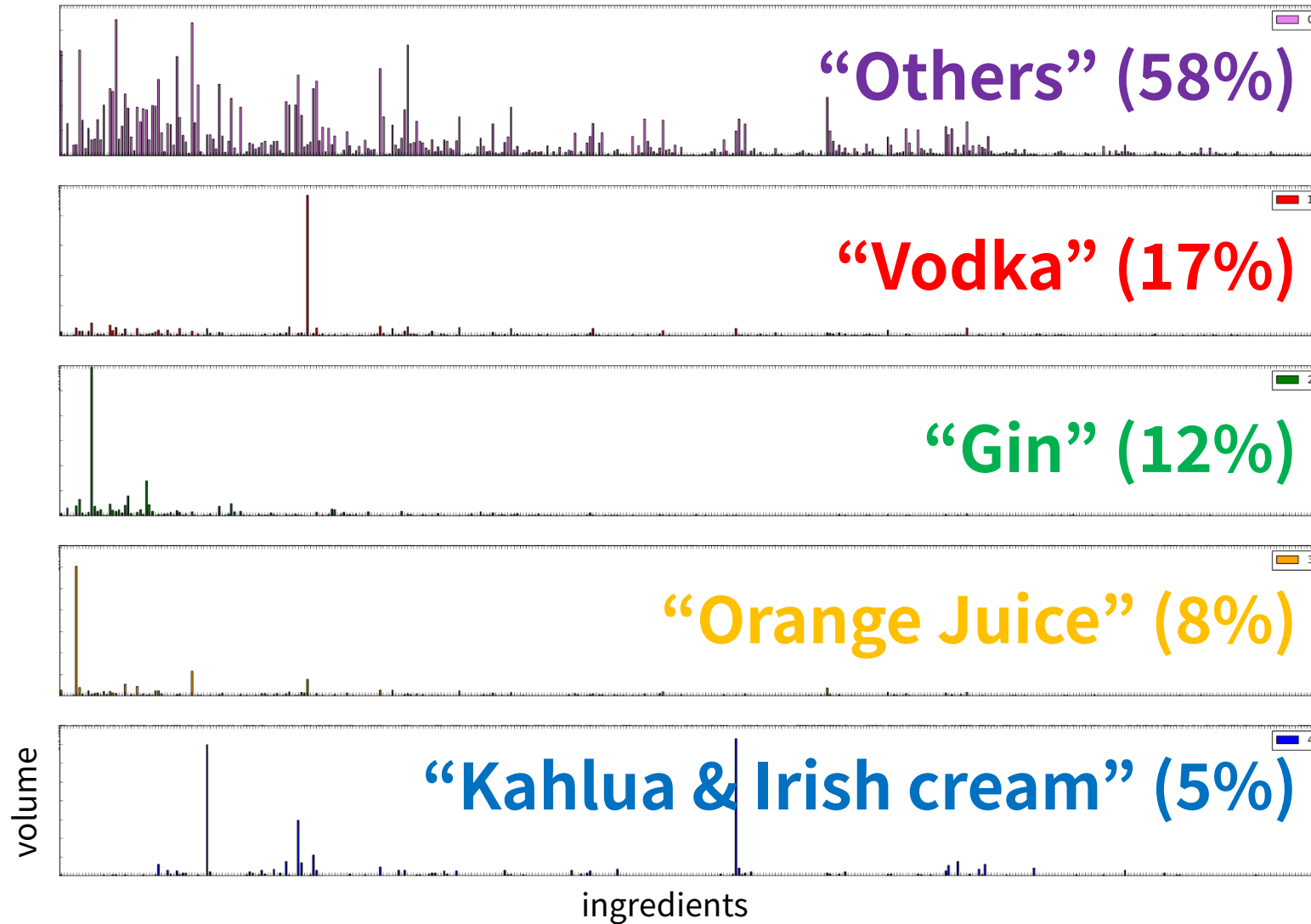
Output: Cape Codder
(score=0.75)



Slide#3: PCA and Clustering



Slide#4: Market Segmentation



Thank you!